

Edge-Enabled UAV Swarm Deployment for Rapid Post-Disaster Search and Rescue

Alaa Awad Abdellatif, Helder Fontes, Andre Coelho, Luis M. Pessoa, Rui Campos
INESC TEC and Faculdade de Engenharia, Universidade do Porto, Portugal

Abstract—This paper presents an optimized framework for Post-Disaster Search and Rescue (PDSR) that leverages multiple Unmanned Aerial Vehicles (UAVs) equipped with integrated radar and communication capabilities to simultaneously address sensing and connectivity requirements. The proposed solution includes a scalable system architecture and an optimization strategy that enable the rapid deployment of UAV swarms with diverse sensing, communication, and edge-enabled coordination features, ensuring enhanced performance in real-world disaster environments. The proposed approach formulates and solves a 3D UAV positioning and power allocation problem to maximize sensing performance and communication efficiency over multiple targets in designated zones. Due to the NP-hard and combinatorial nature of the problem, we propose a Distributed Joint Radar-Communication (DJRC) solution. This solution employs an efficient reward for potential actions and consistently selects the best action that maximizes the reward while ensuring both communications and sensing performance. Simulation results demonstrate significant performance improvements of the proposed solution over state-of-the-art radar- or communication-centric methods, with polynomial complexity dependent on the number of UAVs and linear dependence on the iteration count.

Index Terms—Multi-UAV System, 3D positioning, resource allocation, cooperative detection, power control.

I. INTRODUCTION

In Post-Disaster Search and Rescue (PDSR), Joint Radar-Communication (JRC) systems offer a powerful means to unify high-accuracy sensing with reliable wireless connectivity. This aligns with the emerging vision of “network as a sensor”, where communication infrastructure supports sensing tasks [1]. Unmanned Aerial Vehicles (UAVs) equipped with integrated radar and communication capabilities are ideal for such operations, enabling efficient data collection and transmission in dynamic environments like disaster zones [2], [3]. For instance, UAVs equipped with high-frequency radar offer key advantages over cameras in certain scenarios due to their unique capabilities. Radar enables all-weather operation, functioning reliably in fog, rain, snow, and low-light conditions. It directly measures object velocity for precise motion tracking, penetrates visual obstructions like fog/rain, and remains unaffected by lighting changes. Also, radar’s extended detection range makes it ideal for early warning and wide-area surveillance applications.

Despite the potential and advancements of multi-UAV JRC systems, real-world deployment is complicated by factors such as limited fleet size, restricted coverage area per UAV, and challenging PDSR conditions [4]. These limitations make effective UAV deployment strategy critical for maximizing radar sensory and communications performance. Existing research has explored various aspects of JRC systems, including radar-communication spectrum sharing, trajectory planning and power control, adaptive beamforming and waveform diversity, and distributed systems where multiple transmitters and receivers collaboratively perform radar and communications tasks [5]–[7]. The

authors in [8] developed a beamformer for MIMO dual-function radar-communication system under radar direction mismatch and imperfect channel conditions, employing a max-min optimization solved via semidefinite programming. In [9], a trajectory planning study is presented to optimize UAV movements while improving radar coverage and communications quality. The work in [10], [11] investigated power control strategies, focusing on allocating power between radar and communications subsystems to balance sensing quality and communications throughput, thereby enhancing overall JRC system performance. In [12], a UAV trajectory optimization algorithm for ISAC networks enhances both communication and localization accuracy through a Cramér-Rao bound-based objective and convex optimization. In disaster scenarios, [13] introduces a dual-scheme framework combining satellite-assisted and ISAC-assisted localization, optimizing UAV trajectories and beamforming for improved energy efficiency in Space-Air-Ground Integrated Networks (SAGIN). However, existing approaches primarily rely on centralized or suboptimal solutions, lacking flexibility, scalability, and fault tolerance. Moreover, most learning-based methods employ discrete action spaces, resulting in inefficient optimization strategies, or assume fixed UAV altitudes and stationary ground base stations [10], thus constraining system performance and limiting achievable optimality.

This work addresses the gap in resilient PDSR connectivity by proposing a multi-tiered UAV architecture integrated with a Distributed JRC-based (DJRC) solution. The architecture enables robust coordination among UAV swarms. The DJRC solution jointly optimizes UAVs’ 3D positions and power allocation to maximize sensing quality across all UAVs while maintaining communication performance. We employ a swarm of UAVs organized in a hierarchical architecture to enable scalable sensing and communication operations. High-altitude UAVs function as Flying Base Stations (FBSs), providing wide-area coverage, while low-altitude UAVs, equipped with dual-functional radar-communication systems, perform target-specific sensing and relay collected data to the FBS. This approach ensures rapid and broad coverage of designated regions, efficient resource utilization, and minimal redundancy. Our main contributions are:

- **Trajectory planning and power control:** We formulate an optimization problem for the proposed multi-tiered scenario to jointly optimize UAV locations in 3D space and power allocation between radar and communication functions. The objective is to maximize sensing quality while ensuring communication quality requirements are met.
- **Distributed optimization:** To address the NP-hard and combinatorial complexity of the problem, we propose an edge-based distributed solution that leverages optimization decomposition and an optimality-guided reward mechanism. This approach enables each UAV to independently update its 3D position and power while

This work was carried out within the scope of the HURRICANE project, which has received funding from the European Union’s Horizon Europe research and innovation programme under Grant Agreement 101168017.

jointly balancing sensing accuracy and communication performance across all assigned targets.

- **Improved detection quality and reduced complexity:** Our approach demonstrates superior detection quality compared to traditional methods while reducing computational complexity. This efficiency makes it practical for implementation in PDSR operations.

The following sections present our system model and problem formulation (Section II), the proposed distributed JRC solution (Section III), the performance evaluation results (Section IV), and the conclusions (Section V).

II. SYSTEM MODEL AND PROBLEM FORMULATION

This section introduces the system model, defines key performance metrics for sensing and communication models, and formulates the multi-UAV optimization problem.

A. Multi-UAV JRC System Model

The PDSR multi-UAV system, depicted in Figure 1, is designed to conduct efficient aerial surveys, enhance sensing operations, and forward collected data to a First Responder Center (FRC) for analysis and decision-making. This system integrates two primary components: a high-altitude UAV acting as a FBS, and a swarm of M Low-Altitude (LA) UAVs equipped with dual radar and communications capabilities. These LA-UAVs perform sensing operations for N targets (identified by FBS) over T consecutive time slots, and transmit the sensed data to the FBS, utilizing separate hardware and distinct signals for sensing and communication [14]. The FBS serves two key functions: (i) it performs wide-area, high-altitude imaging to detect target locations, such as collapsed buildings or injured individuals, and assigns them to LA-UAVs for detailed sensing; (ii) it aggregates data from all associated LA-UAVs for transmission to the FRC. The latter serves as the central command for UAV operations, overseeing deployment, coordinating tasks, and monitoring UAV status.

The proposed multi-tiered UAV architecture enables LA-UAVs operations for high-resolution sensing, while the FBS operates at a higher altitude to provide broad surveillance and enhance communications via improved Line-of-Sight (LoS). However, lower UAV altitudes limit individual coverage areas, necessitating more UAVs for wide-area sensing. Conversely, excessively high FBS altitudes may incur increased path loss and signal degradation due to environmental factors. These tradeoffs underscore the importance of carefully designing UAV assignment and coordination strategies to ensure efficient resource utilization, balancing sensing quality with reliable communication performance.

Once optimal LA-UAVs locations are determined, each UAV autonomously navigates to its designated position to conduct continuous sensing and communication tasks. It is assumed that UAVs are equipped with autonomous navigation and obstacle avoidance capabilities, enabling them to follow the safest, most efficient routes, thereby minimizing travel time and enhancing mission effectiveness. The design of backhaul communication protocols for the FBS and the energy consumption due to UAV mobility are beyond the scope of this article, as our primary focus is on optimizing sensing and communication performance; these aspects can be addressed in complementary studies without impacting the validity of our proposed framework.

The coordinates of the m -th UAV at time t are represented as $(x_m(t), y_m(t), H_m(t))$, determined via GPS. The

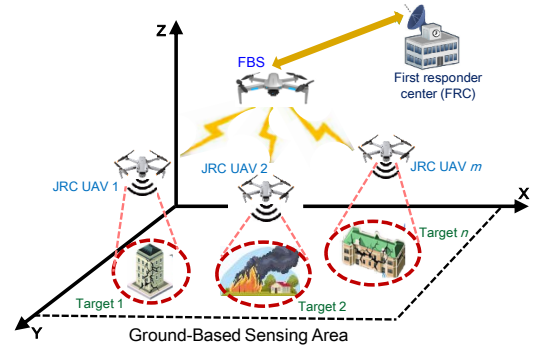


Fig. 1: The considered multi-UAV JRC system model.

coordinates of the FBS are denoted as $(x_h(t), y_h(t), H_h(t))$, where $H_h(t) > H_m(t)$. The coordinates of the n -th target are given by $(x_n, y_n, 0)$. Target locations are identified at the beginning of the mission via initial wide-area scans conducted by high-altitude UAVs. The UAV-target assignments are then defined by the FRC, with each UAV designated to detect a single target at time slot t .

B. Radar Sensing Model

We adopt the Radio Frequency (RF) radar model for sensing due to its robustness in low-visibility conditions (e.g., fog, rain, and darkness) and its superior long-range detection and velocity estimation. Its seamless integration with communication systems makes it well-suited for UAV-based JRC applications. To measure the radar detection quality, we consider a monostatic radar setup¹. The signal-to-noise ratio (SNR) between the m -th UAV and n -th target at time slot t is given by [10]:

$$\eta_{mn}(t) = \frac{p_m^r(t) g_m^T g_m^R \lambda^2 \sigma_n}{(4\pi)^3 \Gamma B_m^r d_{mn}^4(t)}, \quad (1)$$

where p_m^r and B_m^r is the allocated power and bandwidth (BW) for the radar of the m -th UAV, respectively, g_m^T and g_m^R refers to the transmitting and receiving antenna gains, respectively, $\lambda = C/f_c$ is the operating wavelength, and σ_n is the Radar Cross Section (RCS) of the n -target. For high-frequency radars, $B_m^r \ll f_c$, where f_c represents the operating frequency, ensuring effective signal modulation and detection. $\Gamma = kT_0Fl$, where k is the Boltzmann constant, T_0 is the effective noise temperature in terms of thermodynamic temperature, F and l are the radar noise figure and probing loss, respectively. The distance between the m -th UAV and the n -th target, denoted as d_{mn} , is computed using the 3D Euclidean distance formula:

$$d_{mn} = \sqrt{(x_m - x_n)^2 + (y_m - y_n)^2 + (H_m)^2}. \quad (2)$$

To guarantee the detection quality of the m -th UAV for the n -th target, the radar SNR must satisfy: $\eta_{mn}(t) \geq \eta_{\min}$, where $\eta_{\min}$ denotes the minimum required SNR for successful target detection. Accordingly, the maximum detectable range of UAV m to detect target n in time slot t , known as the radar range, is defined as:

$$R_{mn}(t) = \left(\frac{p_m^r(t) g_m^T g_m^R \lambda^2 \sigma_n}{(4\pi)^3 \Gamma B_m^r \eta_{\min}} \right)^{1/4}. \quad (3)$$

This sets a constraint on the m -th UAV to guarantee the detection quality, i.e., $d_{mn}(t) \leq R_{mn}(t)$.

¹In monostatic radar, the range from the target to the transmitter and receiver is identical.

C. Communication Model

While sensing the targets, each UAV needs to transmit sensed data to the FBS. Following the communication model in [10], [15]–[17], the average channel power gain between the m -th UAV and the FBS is calculated by:

$$h_{mh}(t) = K_0^{-1} d_{m0}(t)^{-2} [\xi_m^{\text{LoS}}(t) \mu^{\text{LoS}} + \xi_m^{\text{NLoS}}(t) \mu^{\text{NLoS}}]^{-1}, \quad (4)$$

where $K_0 = (\frac{4\pi f_c}{C})^2$, μ^{LoS} and μ^{NLoS} are the attenuation factors for LoS and Non-Line-of-Sight (NLoS) links, respectively. $\xi_m^{\text{LoS}}(t)$ and $\xi_m^{\text{NLoS}}(t)$ are the LoS and NLoS probabilities between the m -th UAV and the FBS, respectively. The distance between the m -th UAV and the FBS at time slot t is denoted as $d_{mh}(t)$. Based on the allocated transmit power for communication $p_m^c(t)$ and channel gain, the data rate of the m -th UAV on the communication link is calculated by: $r_m(t) = B_m^c \log_2(1 + \Psi_{mh}(t))$, where $\Psi_{mh}(t) = (p_m^c(t) \cdot g_m^T \cdot g_h^R \cdot h_{mh}(t)) / (B_m^c \cdot \delta_0)$ is the signal-to-noise-ratio between the m -th UAV and the FBS, B_m^c is the allocated BW for communication, and δ_0 is the thermal noise [10]. To ensure the quality of data transmission between the UAV and the FBS, we set a threshold $R_{\min}$ for the transmission rate of each UAV, i.e., $r_m(t) \geq R_{\min}$. Alternative SNR or channel models can be seamlessly integrated into our framework without affecting the generality or applicability of the proposed solution methodology, which remains robust under various modeling assumptions.

D. Problem Formulation

In the considered scenario, a swarm of $\mathcal{M}$ JRC-enabled UAVs is deployed to detect targets within a specified region and transmit the data to a FBS. The objective is to maximize the overall radar detection quality while ensuring detection range and communication data rates with the FBS. This requires optimizing each UAV's 3D position to balance radar coverage and communication efficiency, along with its power splitting factor to allocate energy between sensing and communication tasks effectively. This scenario is mathematically formulated as an optimization problem to address the complex trade-offs and derive an efficient strategy for the JRC-enabled UAV network, as detailed below:

$$\mathbf{P}: \quad \max_{\gamma, \mathbf{q}, \mathbf{L}^h} \sum_{i=1}^M w_i \cdot \eta_{mn}(t) \quad (5)$$

subject to:

$$d_{mn}(t) \leq R_{mn}(t), \quad \forall m \in \mathcal{M}, \quad (6)$$

$$r_m(t) \geq R_{\min}, \quad \forall m \in \mathcal{M}, \quad (7)$$

$$d_{mm'}(t) \geq d_g, \quad \forall m \in \mathcal{M} \wedge m \neq m', \quad (8)$$

$$X_{\min} \leq x_m(t), x_h(t) \leq X_{\max}, \quad \forall m \in \mathcal{M}, \quad (9)$$

$$Y_{\min} \leq y_m(t), y_h(t) \leq Y_{\max}, \quad \forall m \in \mathcal{M}, \quad (10)$$

$$0 < H_m, H_h \leq H_{\max}, \quad (11)$$

$$\gamma_m(t) \in [0, 1], \quad \forall m \in \mathcal{M}, \quad (12)$$

where w_i is a weighting coefficient used to assign different priorities to the UAVs (e.g., based on the importance of the targets), satisfying the condition $\sum_{i=1}^M w_i = 1$, and $\gamma_m(t) \in [0, 1]$ is the power split factor for the m -th UAV at time slot t . Hence, the power allocated by the m -th UAV for sensing at time slot t is given by $p_m^r(t) = (1 - \gamma_m(t)) \cdot p_t$, and the power allocated for communication is $p_m^c(t) = \gamma_m(t) \cdot p_t$, where p_t is the total power of each UAV.

The formulated problem in $\mathbf{P}$ aims to determine the optimal locations of the UAVs $\mathbf{q}_m$ and the FBS $\mathbf{L}^h$ to maximize the detection quality across all targets. Here, $\mathbf{q}_m = (x_m, y_m, H_m) \mid m = \{1, 2, \dots, M\}$ represents the location of m -th UAV, and $\mathbf{L}^h = (x_h, y_h, H_h)$ represents the location of the FBS, both defined in a 3D space. Thus, we aim to determine $\mathbf{p}_m$, $\mathbf{L}^h$, and the power splitting factor γ , while maximizing their detection quality and maintaining the required communication data rates with the FBS. The constraint in (6) ensures the distance between the m -th UAV and the n -th target is within the radar range. The constraint in (7) ensures the minimum transmission data rate for all UAVs is met. The constraint in (8) maintains a safe distance d_g between UAVs to prevent collisions, where $d_{mm'}(t)$ is the distance between the m -th UAV and the m' -th UAV. The constraints in (9), (10), and (11) guarantee that all UAV flights are restricted to a region defined by $[X_{\min}, X_{\max}] \times [Y_{\min}, Y_{\max}] \times [H_{\min}, H_{\max}]$. Finally, the constraint in (12) ensures that the power splitting factor γ_m for each UAV m remains within the valid range of $[0, 1]$. Although we do not explicitly model them here to maintain clarity and focus, per-UAV energy and flight-time constraints can be seamlessly incorporated as linear bounds on each UAV's total power and time budgets—requiring only straightforward additional inequalities and preserving both the problem structure and computational complexity.

The formulated problem in $\mathbf{P}$ is NP-hard due to its non-convex and combinatorial nature [18], as evidenced by constraints (7) and (8). Therefore, in the following section, we propose a Distributed Joint Radar and Communication (DJRC) optimization algorithm to address this challenge.

III. DISTRIBUTED JRC SOLUTION

To solve $\mathbf{P}$, we first decompose the problem into manageable sub-problems, which are then solved iteratively using the Distributed Joint Radar-Communication (DJRC) algorithm until convergence. The details of the problem decomposition and DJRC algorithm are provided below.

A. Problem Decomposition

In order to analytically solve the problem in $\mathbf{P}$, we decompose it into two subproblems, each dependent on one or more decision variables and solvable independently [19]. The challenge arises from the coupling of optimization variables (i.e., $\mathbf{q}$, $\mathbf{L}^h$). To address this, we optimize the UAV variables (γ and $\mathbf{q}$) and the FBS variables ($\mathbf{L}^h$) separately. UAV variables are locally optimized at the edge in a distributed manner, while FBS adapting its variables based on the UAVs' updated states without centralized coordination. Thus, we decompose the problem into FBS and UAV subproblems as follows:

$$\mathbf{SP1}: \quad \max_{\mathbf{L}^h} \sum_{i=1}^M w_i \cdot \eta_{mn}(t) \quad (13)$$

subject to:

$$r_m(t) \geq R_{\min}, \quad \forall m \in \mathcal{M}, \quad (14)$$

$$X_{\min} \leq x_h(t) \leq X_{\max}, \quad \forall m \in \mathcal{M}, \quad (15)$$

$$Y_{\min} \leq y_h(t) \leq Y_{\max}, \quad \forall m \in \mathcal{M}, \quad (16)$$

$$0 < H_h \leq H_{\max}, \quad (17)$$

$$\mathbf{SP2}: \quad \max_{\gamma, \mathbf{q}} \sum_{i=1}^M w_i \cdot \eta_{mn}(t) \quad (18)$$

subject to: (6), (7), (8), (9), (10), (11), (12).

B. FBS Optimization

By analyzing **SP1**, we observe that the optimization variable $\mathbf{L}^h$ does not influence the objective $\sum_{i=1}^M w_i \cdot \eta_{mn}(t)$. However, it does affect the constraint in (14). The location of the FBS impacts the data rates (r_m) of the UAVs, but not their detection performance, which depends solely on their location and power. Therefore, the optimal location for the FBS is the one that maximizes the data rates of all UAVs. As a result, the problem in **SP1** is reformulated as follows:

$$\begin{aligned} \mathbf{P2}: \quad & \max_{\mathbf{L}^h} \sum_{i=1}^M r_m(t) \\ & \text{subject to: (15), (16), (17),} \end{aligned} \quad (19) \quad (20)$$

To solve **P2** and determine the optimal location for the FBS, $\mathbf{L}^h$, we employ a gradient-based method, as follows:

1) *Initialization*: To ensure balanced data rates between all UAVs and the FBS, the initial location of the FBS, denoted as $\mathbf{L}_{in}^h$, is determined as follows:

$$\mathbf{L}_{in}^h = \left(\sum_{n=1}^N \frac{x_n}{N}, \sum_{n=1}^N \frac{y_n}{N}, \max(H_m) + d_h \right), \quad (21)$$

where d_h represents the minimum height difference between the maximum UAVs' height and the FBS to avoid collisions. This initial position represents the centroid of all target locations in the horizontal plane and at an altitude greater than the maximum UAV height by a safe margin.

2) *Gradient Calculation*: Compute the gradients of the objective function in (19), i.e., $\nabla f(\mathbf{L}^h)$, where $f(\mathbf{L}^h) = \sum_{i=1}^M r_m(t)$, by taking the partial derivatives of $f(\mathbf{L}^h)$ with respect to each coordinate of $\mathbf{L}^h$, such that:

$$\nabla f(\mathbf{L}^h) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial h} \right), \quad (22)$$

3) *Update Rule*: Update the FBS location $\mathbf{L}^h$ iteratively by using the gradient ascent rule, as follows:

$$\mathbf{L}^h(t+1) = \mathbf{L}^h(t) + \alpha \nabla f(\mathbf{L}^h), \quad (23)$$

where $\alpha > 0$ is the learning rate. Then, the updated location $\mathbf{L}^h(t+1)$ is projected onto the feasible region to ensure it satisfies all constraints in (15), (16), (17).

4) *Termination Criteria*: Repeat steps 2-3 until convergence constraint is satisfied, i.e., $\|\nabla f(\mathbf{L}^h)\| < \epsilon$, where ϵ is a predefined tolerance.

Our gradient-based solution provides a practical and computationally efficient solution for FBS optimization, as it leverages the local gradient to guide the search for the optimal solution. This approach enables faster convergence compared to exhaustive search methods, which is particularly advantageous in large-scale 3D search spaces.

C. UAV Optimization

To maximize the objective function in **SP2**, each UAV should be as close as possible to its assigned target while utilizing the maximum available power for sensing. However, this must be done while satisfying the constraints in (6)-(12). To obtain the optimal solution, each UAV m is initially positioned at the closest safe point to its target, i.e., $x_m = x_n$, $y_m = y_n$, $H_m = d_g$, considering a safe distance d_g , while allocating its entire power to sensing ($\gamma_m = 0$). If the initial configuration violates any constraints, the UAV can adjust its optimization variables by either (i) increasing γ_m to balance sensing and communications power, or (ii)

moving towards the FBS while maintaining the shortest possible distance to both the target and the FBS; this is achieved by moving toward the FBS along a spherical surface centered at the target and incrementally increasing the sphere's radius by a small step Δ_r .

To mathematically formulate our solution, we define a reward function $\mathcal{R}(m)$ for each UAV. This reward function accounts for the increase in achieved rate r_m and the decrease in η_{mn} , at each time step t , due to the taken action by a UAV. Hence, it is defined as:

$$\mathcal{R}(m) = \left(\frac{r_m^{t+1} - r_m^t}{r_m^t} \right) - \left(\frac{\eta_{mn}^t - \eta_{mn}^{t+1}}{\eta_{mn}^t} \right) \quad (24)$$

At each time step t , each UAV takes the action $A_m \in \{a_1, a_2\}$ that maximizes its reward $\mathcal{R}(m)$, such that a_1 refers to increasing communications power, and a_2 refers to moving toward the FBS with step Δ_r , which is formulated as follows:

$$A_m(t) = \arg \max_{A \in \{a_1, a_2\}} \mathcal{R}(m)$$

with the following update rules:

$$\begin{cases} \gamma_m(t+1) = \gamma_m(t) + \Delta_\gamma, & \text{if } A_m(t) = a_1 \\ \mathbf{q}_m(t+1) = \mathbf{q}_m(t) + \Delta_r \cdot \frac{\mathbf{L}^h - \mathbf{q}_m(t)}{\|\mathbf{L}^h - \mathbf{q}_m(t)\|}, & \text{if } A_m(t) = a_2 \end{cases} \quad (25)$$

By using this formulation, the UAVs independently adjust their position and power splitting based on reward maximization. This iterative process ensures the UAVs select the best action, update their positions or power levels accordingly, and remain as close as possible to their target while moving toward the FBS. The process continues until all constraints in (6)-(12) are satisfied. We argue that as long as the step sizes (Δ_γ and Δ_r) are sufficiently small, the system gradually converges to the best configuration that maximizes the objective function in **SP2**, while ensuring all constraints are satisfied.

D. Distributed Joint Radar-Communications (DJRC)

This section introduces DJRC, a distributed and iterative algorithm for optimal UAV positioning and power allocation. Building on the problem decomposition discussed in the previous subsections, DJRC is designed to efficiently solve the problem in **P**, in practical scenarios where each UAV is aware only of its assigned target location.

According to DJRC, each UAV is initially positioned as close as possible to its assigned target while utilizing its total power for sensing. To ensure balanced data rates among UAVs, the FBS is initially placed at the centroid of all target locations in the horizontal plane, with an altitude exceeding the maximum UAV height by a safe margin, as defined in (21). Then, at each iteration, each UAV m evaluates its reward function in (24) to determine whether to increase its power split factor γ_m by Δ_γ or move toward the FBS along the shortest path, as defined in (25). Once all UAVs have taken their respective actions independently, the FBS solves its optimization problem in **P2** to update its location, based on the updated locations of LA-UAVs. This iterative process continues until convergence is achieved—i.e., all constraints in (6)-(12) are satisfied—or until a predefined maximum number of iterations T_m is reached. The main steps of the DJRC algorithm are presented in Algorithm 1.

The computational complexity of the DJRC algorithm is driven by iterative UAV updates and FBS location optimization. In each iteration, the reward computation and

Algorithm 1 DJRC Algorithm

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1: Initialization:  $x_m = x_n$ ,  $y_m = y_n$ ,  $H_m = d_g$ , and  $\gamma_m = 0$ ,
    $\forall m \in \mathcal{M}$ .
2: Compute the initial FBS position using Equation (21).
3: for  $t = 1$  to  $T_m$  do
4:   for  $m = 1$  to  $M$  (in parallel) do
5:     Compute reward function  $\mathcal{R}(m)$  using (24).
6:     Select action  $A_m$  that maximizes  $\mathcal{R}(m)$ :
7:     if  $A_m = a_1$  then
8:       Increase power split factor using (25).
9:     else if  $A_m = a_2$  then
10:      Move toward FBS using (25).
11:     end if
12:   end for
13: FBS Update: Solve problem P2 to update the FBS location  $\mathbf{L}^h$ .
14: if All constraints in (6)-(12) are satisfied then
15:   Break
16: end if
17: end for
18: Output: Optimal UAV positions  $\mathbf{q}_m$  and power split factors  $\gamma_m$ , and
   FBS location  $\mathbf{L}^h$ .

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action selection for each UAV require $O(1)$ operations, leading to $O(M)$ complexity across all UAVs. The FBS location is optimized via a gradient-based method to solve **P2**, with a complexity of $O(T_F)$, where T_F denotes the number of inner iterations. Given that the outer loop runs for at most T_m iterations, the overall complexity becomes $O(T_m \cdot T_F \cdot M)$, reflecting polynomial growth with the number of UAVs and linear dependence on iteration counts. While this complexity assumes centralized execution, our decentralized design distributes the computation among UAVs, reducing the per-UAV complexity to $O(T_m \cdot T_F)$.

IV. PERFORMANCE EVALUATION

In this section, we first demonstrate the convergence of our solution. Then, we compare its performance against two state-of-the-art methods: Fixed Radar Optimized Communications (FROC) [20] and Optimized Radar Fixed Communications (ORFC). The FROC method optimizes communication throughput across all UAVs while assuming a fixed radar range, neglecting the joint radar-communication nature of the system [20]. In contrast, the ORFC method prioritizes radar performance by positioning UAVs to satisfy minimum data rate constraints, assuming fixed UAV altitude and FBS location, as in [10]. Both approaches use fixed power allocation and are evaluated under identical environmental conditions for fair comparison with our solution.

Our performance evaluation is conducted under two scenarios: (i) varying the number of targets, and (ii) varying the total available power at each UAV. In each scenario, we evaluate and compare the radar SNR and the overall data rates across all UAVs for each method. The main simulation parameters are summarized in Table I, and all simulations were carried out in MATLAB.

First, the convergence of the proposed DJRC algorithm is illustrated in Figures 2 and 3. These figures consider three UAVs monitoring three targets, supported by an FBS. Figure 2 depicts the 3D trajectory of each UAV while executing DJRC. Each UAV begins at the nearest feasible point to its assigned target and iteratively advances toward the FBS, adjusting its power-split ratio to balance sensing and communication. This process repeats until all UAVs converge to positions that optimize the sensing–communication trade-off while satisfying every constraint.

Figure 3 further illustrates the tradeoff between sensing and communication by showing the evolution of radar

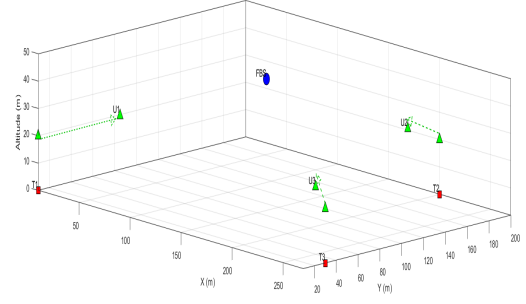
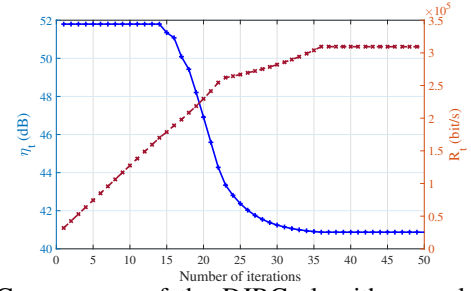


Fig. 2: 3D UAV locations during the execution of the DJRC.

Fig. 3: Convergence of the DJRC algorithm: evolution of total received SNR (η_t) and sum data rate (R_t).

performance—measured by the weighted sum of radar SNR across all UAVs (i.e., $\eta_t = \sum_{i=1}^M w_i \cdot \eta_{mn}$)—and communication quality, represented by the sum of data rates (R_t), as the DJRC algorithm iteratively converges. Initially, with UAVs positioned at the closest feasible points to their targets, the system achieves maximum sensing performance at the cost of lower communication quality. To balance this tradeoff, UAVs gradually adjust their power split factors and move toward the FBS, strategically located at the centroid of all UAVs. This process continues until all UAVs converge to optimal positions that satisfy the system constraints while maximizing overall performance. This adaptive process leads the system to an optimal configuration, balancing sensing and communication objectives. Remarkably, convergence is reached within 40 iterations, demonstrating the efficiency of DJRC. The solution avoids exhaustive search by guiding UAVs along the shortest reward-based paths, achieving polynomial-time complexity. This makes DJRC highly suitable for time-sensitive post-disaster search and rescue missions, where rapid convergence and real-time performance are critical.

For comparison with the FROC and ORFC, we consider two scenarios: (i) varying the number of targets, and (ii)

TABLE I: Simulation Parameters.

Parameter	Value	Parameter	Value
$X_{\min}$	0 m	μ_{NLoS}	2
$X_{\max}$	1000 m	g_m^r	20
d_g	40 m	g_h^r	20
p_t	30 W	σ_n	1 m ²
B_m^c	40 MHz	f_c	5 GHz
B_m^r	20 MHz	C	3×10^8 m/s
$Y_{\min}$	0 m	$R_{\min}$	0.1 Mbit/s
$Y_{\max}$	1000 m	δ_0	0.5×10^{-10} W/Hz
$H_{\max}$	100 m	k	1.38×10^{-23} J/K
$\xi_{m\text{LoS}}$	0.95	T_0	290 K
$\xi_{m\text{NLoS}}$	0.5	F	5 dB
μ_{LoS}	0.5	l	0.8

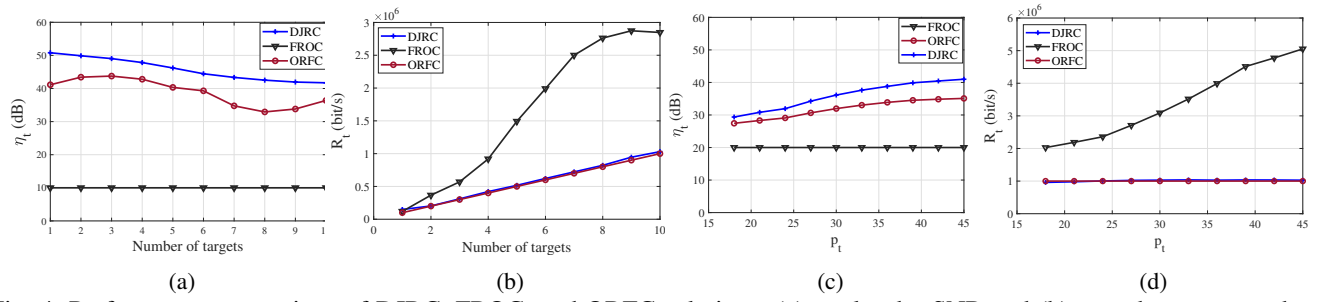


Fig. 4: Performance comparison of DJRC, FROC, and ORFC solutions: (a) total radar SNR and (b) sum data rates, under varying numbers of targets; (c) η_t , and (d) sum data rates, under varying total transmit power per UAV.

varying the total power at each UAV (see Figure 4). The first scenario assesses the performance of each method as the complexity of the target distribution increases. Figure 4 demonstrates that DJRC solution consistently provides the best overall performance by maintaining the highest detection quality η_t while satisfying the data rate constraint. As the number of targets increases, the resources allocated to each UAV decrease, leading to a reduction in detection quality, as in Figure 4-(a). Nevertheless, the DJRC algorithm continues to outperform the FROC and ORFC solutions because it jointly optimizes both radar and communications functionalities. In contrast, the FROC focuses solely on communications [20], leading to UAVs being allocated at the maximum radar range, which results in lower detection quality ($\eta_{\min}$) while maximizing the sum data rate R_t (see Figure 4-(b)). On the other hand, the ORFC solution prioritizes radar detection quality, allocating UAVs to positions that maximize radar performance while ensuring the data rate constraint. However, it ignores the joint optimization of UAV locations and power split factors.

The second scenario fixed the number of targets at 10 and varied the total available power at each UAV, P_t (see Figure 4-(c)). This setup assesses how the detection quality improves as the available power increases. In Figure 4-(c), the DJRC solution consistently delivers the best overall performance in terms of detection quality, η_t , while ensuring that the total data rate of all UAVs satisfies the 1 Mbit/s constraint. This demonstrates the ability of the DJRC algorithm to effectively balance both radar and communications requirements, outperforming the FROC and ORFC solutions even as power availability increases.

V. CONCLUSION

This paper proposes an efficient, distributed solution for optimizing UAV positioning in post-disaster environments without relying on ground infrastructure. By leveraging UAVs equipped with dual radar-communication capabilities, the system addresses critical challenges such as damaged infrastructure and limited accessibility, enabling rapid detection and data collection in hard-to-reach areas. This is particularly valuable for time-sensitive applications like disaster response. The proposed approach jointly optimizes power splitting and UAV positioning to maximize sensing quality while ensuring reliable communication, outperforming alternative methods. It achieves polynomial computational complexity, making it suitable for real-time deployment. This work also lays the foundation for future research on multi-UAV cooperation, trajectory and resource optimization, and integration of diverse sensing and communication technologies.

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